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COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY (CoICT)

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REGISTRATION NO.: 2024-04-07577

PROGRAMME: BSc. TELECOMMUNICATION ENGINEERING

COURSE CODE: CS 175

COURSE NAME: PROGRAMMING IN JAVA

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TASK: ASSIGNMENT TWO

DATE OF SUBMISSION: 19th June 2025

To prepare a portfolio of all sample program pieces under the following topics:

1. Applets,
2. Array and
3. String.

1. APPLET.

What is an Applet?

- It is a special type of program that is embedded in the webpage to generate the dynamic content.
- Applet is mostly used in games and animation.
- It runs inside the browser and works at client side.

Advantages & Disadvantages;

Advantages

- It works at client side so less response time.
- Secured.
- It can be executed by browsers running under many platforms, including Linux, Windows, Mac, etc.

Disadvantages

- Plugin is required at client browser to execute applet.

SAMPLE PROGRAMS:

```
import java.applet.Applet;
import java.awt.Graphics;
/*
<applet code="DaudiMtale.class" width="300" height="150">
</applet>
*/
   public class DaudiMtale extends Applet {
    public void paint(Graphics g) {
        g.drawString("Hello, this is my first Applet!", 20, 50);
```

```
    }
  }
```

Applet with Initialization and Start Methods:

```java
import java.applet.Applet;
import java.awt.Graphics;

public class DaudiMtale extends Applet {

    public void init() {
        System.out.println("Applet initialized");
    }

    public void start() {
        System.out.println("Applet started");
    }

    public void paint(Graphics g) {
        g.drawString("Applet Lifecycle Demo", 50, 50);
    }

    public void stop() {
        System.out.println("Applet stopped");
    }

    public void destroy() {
        System.out.println("Applet destroyed");
    }
}
```

Passing Parameters to Applet:

```java
import java.applet.Applet;
import java.awt.Graphics;

/*
<applet code="DaudiMtale.class" width="300" height="150">
    <param name="username" value="JohnDoe">
</applet>
*/
```

```java
public class DaudiMtale extends Applet {
    String user;

    public void init() {
        user = getParameter("username");
    }

    public void paint(Graphics g) {
        g.drawString("Welcome, " + user, 50, 50);
    }
}
```

Using Graphics in Applets:

```java
import java.applet.Applet;
import java.awt.Color;
import java.awt.Graphics;

public class DaudiMtale extends Applet {
    public void paint(Graphics g) {
        g.setColor(Color.blue);
        g.drawRect(20, 20, 100, 50);
        g.setColor(Color.red);
        g.fillOval(150, 20, 50, 50);
    }
}
```

2. ARRAY.

Arrays in Java:

• Java array is an object which contains elements of a similar data type.

• The elements of an array are stored in a contiguous memory location.

• It is a data structure where we store similar elements.

• We can store only a fixed set of elements in a Java array.

• Array in Java is index-based, the first element of the array is stored at the 0th index, 2nd element is stored on 1st index and so on.

SAMPLE PROGRAMS:

```
class DaudiMtale{
    public static void main(String[] args){
      int a[]=new int[5];
        a[0]=10;
        a[1]=20;
        a[2]=70;
       a[3]=40;
       a[4]=40;
        for(int i=0;i<a.length;i++){
          System.out.println(a[i]);
        }
    }
}
```

```
class DaudiMtale{
    public static void main(String[] args){
     int a[]={33,3,4,5};
          for(int i=0;i<a.length;i++)
           System.out.println(a[i]);
    }
}
class DaudiMtale{
    public static void main(String[] args){
        int x[ ] = new int [5];
         for(int i = 0; i < x.length; ++i){
         x[i] =1+2;
    }
            for(int i = 0; i < x.length; ++i){
         System.out.println(x[i]);
               }
    }
}
```

```java
class DaudiMtale{
    public static void main(String[] args){
     int[] age = {12, 4, 5, 2, 5};
     System.out.println("Accessing array elements:");
     System.out.println("First element: " + age[0]);
     System.out.println("Second element: " + age[1]);
     System.out.println("Third element: " + age[2]);
     System.out.println("Fourth element: " + age[3]);
     System.out.println("Fifth element: " + age[4]);
    }
}
```

```java
class DaudiMtale{
    public static void main(String[] args){
   int[] age = {12, 4, 5};
   System.out.println("Using for Loop:");
   for(int i = 0; i < age.length; i++) {
     System.out.println(age[i]);
   }
    }
}
```

```java
class DaudiMtale{
    public static void main(String[] args){
   int[] age = {12, 4, 5};
   // loop through the array
   // using for loop
   System.out.println("Using for-each Loop:");
      for(int a : age) {
        System.out.println(a);
   }
    }
}
```

```java
class DaudiMtale{
    public static void main(String[] args){
      int[] numbers = {2, -9, 0, 5, 12, -25, 22, 9, 8, 12};
        int sum = 0;
          Double average;
```

```
       for (int number: numbers) {
          sum += number;
            }
           int arrayLength = numbers.length;
             average =  ((double)sum / (double)arrayLength);
               System.out.println("Sum = " + sum);
                 System.out.println("Average = " + average);
    }
}
```

```
import java.util.ArrayList;
import java.util.Collections;
public class DaudiMtale{
    public static void main(String[] args) {
    ArrayList arrayList = new ArrayList();
    arrayList.add("A");
    arrayList.add("B");
    arrayList.add("C");
    arrayList.add("D");
    arrayList.add("E");
    System.out.println("Before Reverse Order: " + arrayList);
    Collections.reverse(arrayList);
    System.out.println("After Reverse Order: " + arrayList);
   }
}
```

```
 public class DaudiMtale{
  public static void main(String[] args) {
    int[] numbers = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };
      System.out.println("Array before reverse:");
        for (int i = 0; i < numbers.length; i++) {
          System.out.print(numbers[i] + " ");
     }
         for (int i = 0; i < numbers.length / 2; i++) {
           int temp = numbers[i];
             numbers[i] = numbers[numbers.length - 1 - i];
                 numbers[numbers.length - 1 - i] = temp;
    }
                 System.out.println("\nArray after reverse:");
                      for (int i = 0; i < numbers.length; i++) {
```

```
                System.out.print(numbers[i] + " ");
            }
        }
    }
```

3. STRING.

A string is a sequence of characters. In Java, strings are objects.

Strings in Java are immutable, meaning their values cannot be changed once created.

- There are two ways to create strings:

By String Literal

Example: String s =”Hello”

- Stored in the string constant pool (a special area in heap memory).
- If the same value exists, a reference to the existing object is returned.
- No new object is created if the string already exists in the pool.

Why Java Uses String Literals**:**

- To make Java more memory-efficient by avoiding duplicate string objects.

By new Keyword

Example: String s = new String("Welcome");

- Creates a new object in the heap (outside the string pool).
- The literal "Welcome" may also be stored in the string constant pool.
- Results in two objects: one in the heap and one in the pool.

SAMPLE PROGRAMS:

Concatenation Method:

```
public class DaudiMtale{
  public static void main(String[] args) {
   String s="Sachin";
    s=s.concat(" Tendulkar");
     System.out.println(s); //Sachin Tendulkar

  }
}
```

Comparison Method:

```
public class DaudiMtale{
   public static void main(String[] args) {
   String s1="Sachin";
   String s2="Sachin";
   String s3=new String("Sachin");
   String s4="Saurav";

  System.out.println(s1.equals(s2));//true
  System.out.println(s1.equals(s3));//true
  System.out.println(s1.equals(s4));//false
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
    String s1="Sachin";
  String s2="SACHIN";

  System.out.println(s1.equals(s2));//false
  System.out.println(s1.equalsIgnoreCase(s3));//true
  }
}
```

```
public class DaudiMtale{
 public static void main(String[] args) {
 String s1="Sachin";
 String s2="Sachin";
 String s3=new String("Sachin");
  System.out.println(s1==s2);//true (because both refer to same instance)
   System.out.println(s1==s3);/*false(because s3 refers to instance created in
nonpool)*/
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
  String s1="Sachin";
  String s2="Sachin";
  String s3="Ratan";
  System.out.println(s1.compareTo(s2));//0
  System.out.println(s1.compareTo(s3));//1(because s1>s3)
  System.out.println(s3.compareTo(s1));//-1(because s 3 < s1 )
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
  String s="Sachin"+" Tendulkar";
  System.Out.println(s);//Sachin Tendulkar
  String s1=50+30+"Sachin"+40+40;
  System.Out.println(s1);//80Sachin4040
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
  String s1="Sachin ";
  String s2="Tendulkar";
  String s3=s1.concat(s2);
  System.out.println(s3);//Sachin Tendulkar
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
   String s="Sachin Tendulkar";
   System.out.println(s.substring(6));//Tendulkar
   System.out.println(s.substring(0,6));//Sachin
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
  String s="Sachin";
  System.out.println(s.toUpperCase());//SACHIN
  System.out.println(s.toLowerCase());//sachin
  System.out.println(s);//Sachin(no change in original)
  }
}
```

```
public class DaudiMtale{
  public static void main(String[] args) {
  String s="  Sachin  ";
  System.out.println(s);//  Sachin
  System.out.println(s.trim());//Sachin
  }
}
public class DaudiMtale{
  public static void main(String[] args) {
   String s="Sachin";
   System.out.println(s.startsWith("Sa"));//true
   System.out.println(s.endsWith("n"));//true
  }
}
```

```java
public class DaudiMtale{
  public static void main(String[] args) {
  String s="Sachin";
  //charAt() method returns the character at the specified index
  System.out.println(s.charAt(0));//S
  System.out.println(s.charAt(3));//h
  //length() method returns the length of the string
  System.out.println(s.length());//6
  }
}
```

```java
public class DaudiMtale{
  public static void main(String[] args) {
    // replace() method returns a new string from
    // the original string, by replacing all occurrences
    // of oldChar with newChar.
        String word1 = "rare";
        String word2 = word1.replace('r', 'd');
         //word2 is “dade”, but word1 is still "rare”
            System.out.println("word2 is: " + word2);
  }
}
```